Cryogenic Kinetics Models

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This note collects the simple temperature-dependent kinetics models used in the cryogenic extension of the project.

1. Defect migration

A first-order Arrhenius model for defect diffusivity is

$$D_{\text{def}}(T) = D_0 \exp\left(-\frac{E_m}{k_B T}\right).$$

Taking the logarithm gives

$$\ln D_{\text{def}} = \ln D_0 - \frac{E_m}{k_B T}.$$

This predicts an approximately straight line on an Arrhenius plot of $\ln D_{\text{def}}$ versus $1/T$.

2. Trap emission

For thermally activated emission from a defect level, use

$$e(T) = e_0 \exp\left(-\frac{E_a}{k_B T}\right).$$

The corresponding emission time constant is

$$\tau_{\text{emit}}(T) = \frac{1}{e(T)}$$

and, if $e_0 = 1/\tau_0$,

$$\tau_{\text{emit}}(T) = \tau_0 \exp\left(\frac{E_a}{k_B T}\right).$$

Thus the trap-emission timescale grows rapidly as temperature decreases.

3. Annealing

A first-order annealing-rate model is

$$R_{\text{ann}}(T) = R_0 \exp\left(-\frac{E_{\text{ann}}}{k_B T}\right).$$

A simple annealing timescale can be defined as

$$\tau_{\text{ann}}(T) = \frac{1}{R_{\text{ann}}(T)}.$$

This makes clear that defect recovery can become extremely slow in the cryogenic regime.

4. Interpretation

These models are not meant to represent one unique microscopic defect species. Instead, they are first-order phenomenological models used to illustrate how thermally activated kinetics collapse as the system moves toward low temperature.